

QA

Quality Assurance (QA) is a proactive process of managing a product that takes place during and after product development. QA ensures a product is well developed and tested before the final delivery to the consumer. It focuses on prevention rather than cure. QA team makes sure are they building the product right, are it well maintained, are it well intact, are it well put it all together, are it well tested before the deadline and things like that. There are QA not just in the software development field but there are everywhere where a product development is involved. Therefore, it's a very important form of quality management.

QC

Quality Control (QC) is a reactive process of monitoring a product after it is developed after the distribution. It focuses on checking and ensuring the product development to meet the quality standards. It is more about damage control and cure rather than prevention. That is why it is a reactive process. QC provides a product as per customer's needs.

Testing

Testing is a process of executing a program with the intent of finding bugs and errors. It differs from QA because testing focuses only to find errors whereas QA finds errors and resolves it ensuring there is no lack of quality in the system. Testing is also a reactive process. It is product focused instead of process focused.

SDLC

Software Development Lifecycle (SDLC) is a systematic process of developed by software engineers to develop software efficiently with high quality on time. It helps to:

- track progress
- minimize failure
- manage changes

SDLC phases include:

- **Initiation:** It is a phase where a proposal of the system or product is created.
- **System Concept Development:** This phase defines the scope, resources, cost of the system.
- **Planning:** Project Plan is created is created through Gantt Chart, WBS, etc.
- **Requirement analysis:** Functional and non-functional requirement is defined. SRS is also created.
- **Design:** Transformation of requirement into blueprint is done. Decides artchitecure, design pattern, etc.
- **Development:** Actual coding of the system is made.
- **Testing:** Here the main part of the testing is done and this is the phase where QA is involved the most.Things like validation and verification is done.
- **Deployment and Maintenance:** Includes bugs correction or debugging, making slight changes, updates and monitoring.

STLC

Software Testing Lifecycle (STLC) is a process that verifies whether the software quality meets the requirement or not. It is a step by step process that goes through different phases of testing over the time frame.

It's phases include:

- **Requirement Analysis:** This is the first step of STLC. Here, the QA team understand and gather the requirements of what is to be tested. It includes things like reviewing the requirements, gathering information from stakeholders, interviewing them, identifying risk related to the test and so on.
- **Test Planning:** Here, the test cases are defined. Calculation of effort, cost, resources are done. Identifying the scope, objectives and developing a testing strategy is performed. Preparation of Requirement Traceability Matrix (RTM) is done.
- **Test Case Development:** It involves creation and verification of the test cases after the test planning phase. Various test data and scripts is created. RTM is also updated to map the requirements.
- **Test Environment Setup:** It decides the software and hardware component in which the system should be tested. It can be done simultaneously with the Test Case Development phase. Things like:
 - performing smoke tests
 - setting up test environment and test data is done.
- **Test Execution:** The actual implementation of test is done. It is a DO phase of the STLC. The activities include:
 - defect logging
 - test result analysis
 - defect retesting
 - test reporting
- **Test Closure:** It is the last step of the STLC. All the testing cases are well documented. It ensures testing related activities are completed and software is all ready to go. It focuses on final delivery. This phase decides whether to give a product a green light or not. Its activities include:
 - test summary report

- defect tracking
- test setup cleanup
- feedback and improvements if required

Roles and Responsibilities of QA Engineer

QA is a critical term in the development field. The team relies on the QA to give a good product to the customer and satisfy them. QA engineer ensures the software passes through all the phases before the final delivery meeting its quality standards and requirements. QA engineer is involved in various roles like:

- planning
- monitoring
- executing
- controlling
- completing of a product over different stages and time.

QA engineer:

- creates standards and procedure for the team to follow.
- execute manual and automated tests.
- implement the test scripts.
- keep track of the quality of the product.
- communicate with the stakeholders.

These all roles seem so mouthful even for QA engineer. So, they obviously need some additional skillsets like communication skills, interpersonal skills, collaborative among the team members, leadership and managerial ability and so on.

Types of Testing: Manual Testing and Automated Testing

There are mainly two different kinds of testing i.e. manual and automated testing. However, there are many sub-divisions of these types and also the sub-divisions of sub-division.

Manual Testing refers to the testing that is done completely with human effort without any use of automation tools and scripts. Testers make test cases of the code and give the final report. It is time consuming compared to automated testing and there is also high chance of human error as it is done by the people. It is done because automated testing is not enough and in some cases, they cannot be used. This test is flexible and gives very fast feedback to the people. It is done to explore the new features in a system, to check the UI and colors is fit and easy to use, whether the system supports and is compatible or not. This method is further more divided into 3 categories: White Box, Black Box and Grey Box testing method.

Manual testing is:

- Flexible
- Gives fast feedback
- Versatile
- Easy to use and perform
- Less expensive

Automated Testing refers to the testing done by various automation tools and scripts. It does not require human work. This testing demands a high programming and automation skills of the testers to perform. Hence, it is very costly. This method is comparatively less time-consuming than that of manual testing. It is very useful when the program have repetitive tasks. It finds more bugs as it covers very high code. There is no fatigue of the tester to do

this, just initial automation script requires human work. Hence, it is accurate and cost-worthy.

Automated Testing is:

- Accurate
- Fast
- Find more bugs
- More code coverage

Summary: SDLC/STLC

Software Development Lifecycle (SDLC) is a systematic approach that focuses on building the system in a uniform manner efficiently and with high quality on time. This cycle defines how the system passes through each and every phase of development process before it is market ready and ready to use. It ensures the product must be adaptable and be able to change anytime without harming the previous work with the help of different methodologies like agile, spiral and so on. This phase does not overlook any kinds of things starts with proper planning of the project, goes through the analysis, produce and creates design and deploy to the consumer. If the consumer is satisfied, it's good, if not, the product is again maintained accordingly to the needs and demands of the consumer.

Software Testing Lifecycle (STLC) is a systematic and incremental process of testing the system according to the needs and quality standards. It also goes through its own sets of phases like requirement gathering, planning, test case development, execution and closure. It gives green light to the product before the final delivery. It test cases, defines what is to be tested , executes the test scripts and monitor and maintain the test related activities.



